FOR A NOETHERIAN RING THE KOSZUL COMPLEXES OF A SEQUENCE ARE PRO-ISOMORPHIC TO THE QUOTIENTS BY THE POWERS OF THE SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement span includes the author's own footnote explaining the consequence of the pro-isomorphism for the factorizations; the footnote is delivered verbatim. The six references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The criterion for an inverse system of distinguished triangles to induce a pro-isomorphism. invoked at proof lines 27188-27190. Reduces the pro-isomorphism statement about K_n^bullet to pro-zeroness of the truncation system.
- The essentially constant cohomology criterion for a bounded complex. invoked at proof lines 27192-27195. Reduces pro-zeroness of the truncation system to pro-zeroness of the individual cohomology systems.
- The Koszul complex of a sequence, its degree bounds and the annihilation of its cohomology by the ideal generated by the powers. used at proof lines 27191-27192 and 27206-27207. Supplies the degree range and the annihilation that the transition-map computation needs.
- Artin-Rees applied to an ideal and a submodule of a finite module. invoked at proof lines 27209-27211. Is the input that forces the transition map on cohomology to vanish.

Source reference. Original lemma 00IN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 685–685. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0G3B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20086–20086. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0G3C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20087–20087. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08J5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10838–10838. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 6660: sources/stacks-more-algebra/pinned-original-source.tex, source lines 6660–6660. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original definition 26305: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26305–26305. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *In Situation 26305. If A is Noetherian, then the pro-objects $\{K_n^\bullet\}$ and $\{A/(f_1^n, \ldots, f_r^n)\}$ of $D(A)$ are isomorphic*[1]*.*

Proof. We have an inverse system of distinguished triangles

$$\tau_{\leq -1}K_n^\bullet \to K_n^\bullet \to A/(f_1^m, \ldots, f_r^m) \to (\tau_{\leq -1}K_n^\bullet)[1]$$

See Derived Categories, Remark 08J5. By Derived Categories, Lemma 0G3C it suffices to show that the inverse system $\tau_{\leq -1}K_n^\bullet$ is pro-zero. Recall that $K_n^\bullet$ has nonzero terms only in degrees i with $-r \leq i \leq 0$. Thus by Derived Categories, Lemma 0G3B it suffices to show that $H^p(K_n^\bullet)$ is pro-zero for $p \leq -1$. In other words, for every $n \in \mathbf{N}$ we have to show there exists an $m \geq n$ such that $H^p(K_m^\bullet) \to H^p(K_n^\bullet)$ is zero. Since A is Noetherian, we see that

$$H^p(K_n^\bullet) = \frac{\mathrm{Ker}(K_n^p \to K_n^{p+1})}{\mathrm{Im}(K_n^{p-1} \to K_n^p)}$$

is a finite A-module. Moreover, the map $K_m^p \to K_n^p$ is given by a diagonal matrix whose entries are in the ideal $(f_1^{m-n}, \ldots, f_r^{m-n})$ as $p < 0$. Note that $H^p(K_n^\bullet)$ is annihilated by $J = (f_1^n, \ldots, f_r^n)$, see Lemma 6660. Now $(f_1^{m-n}, \ldots, f_r^{m-n}) \subset J^t$ for $m - n \geq tn$. Thus by Algebra, Lemma 00IN (Artin-Rees) applied to the ideal J and the module $M = K_n^p$ with submodule $N = \mathrm{Ker}(K_n^p \to K_n^{p+1})$ for m large enough the image of $K_m^p \to K_n^p$ intersected with $\mathrm{Ker}(K_n^p \to K_n^{p+1})$ is contained in $J\,\mathrm{Ker}(K_n^p \to K_n^{p+1})$. For such m we get the zero map. □

[1] In particular, for every n there exists an $m \geq n$ such that $K_m^\bullet \to K_n^\bullet$ factors through the map $K_m^\bullet \to A/(f_1^m, \ldots, f_r^m)$.